Chapter 1

Introduction

Power Electronics

- Power electronics circuits convert electric power from one form to another using electronic devices.
- Power electronics circuits function by using semiconductor devices as switches, thereby controlling or modifying a voltage or current.
- Applications of power electronics range from high-power conversion equipment, such as DC power transmission, to everyday appliances, such as screwdrivers, computer power supplies, cell phone chargers, and hybrid automobiles.

Typical applications of power electronics include converting AC to DC, converting DC to AC, converting an unregulated DC voltage to a regulated DC voltage, and converting an AC power source from one amplitude and frequency to another.

Power electronics includes applications of circuit theory, control theory, electronics, electromagnetics, microprocessors (for control), and heat transfer.
Advances in semiconductor switching capability, combined with the desire to improve the efficiency and performance of electrical devices, have made power electronics an important and fast-growing area in electrical engineering.

Converter Classification

- The objective of a power electronics circuit is to match the voltage and current requirements of the load to those of the source.
- Power electronics circuits convert one type or level of a voltage or current waveform to another and are hence called *converters.* Converters serve as an interface between the sources and load (Fig. 1-1).

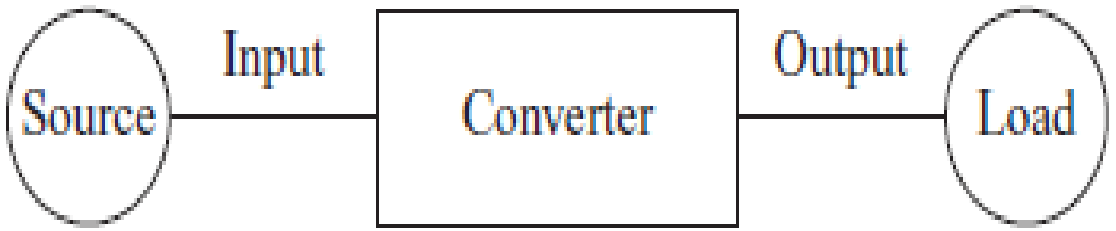


Figure 1-1 A source and load interfaced by a power electronics converter.

Converters are classified by the relationship between input and output:

AC input/DC output

The AC-DC converter produces a DC output from an AC input.
Average power is transferred from an AC source to a DC load.
The AC-DC converter is specifically classified as a *rectifier*.
For example, an AC-DC converter enables integrated circuits to operate from a 60-Hz AC line voltage by converting the AC signal to a DC signal of the appropriate voltage.

DC input/AC output

The DC-AC converter is specifically classified as an *inverter*.

In the inverter, average power flows from the DC side to the AC side.

Examples of inverter applications include producing a $120 - V\ rms\ 60 - Hz$ voltage from a $12 - V$ battery and interfacing an alternative energy source such as an array of solar cells to an electric utility.

DC input/DC output

The DC-DC (Chopper) converter is useful when a load requires a specified (often regulated) DC voltage or current but the source is at a different or unregulated DC value.

For example, 5 V may be obtained from a 12-V source via a DC-DC converter

AC input/AC output

The AC-AC (AC voltage controller) converter may be used to change the level and/or frequency of an AC signal.

Examples include a common light-dimmer circuit and speed control of an induction motor.

Some converter circuits can operate in different modes, depending on the circuit and control parameters.
For example, some rectifier circuits can be operated as inverters by modifying the control on the semiconductor devices.
In such cases, it is the direction of average power flow that determines the converter classification.

In Fig. 1-2, if the battery is charged from the AC power source, the converter is classified as a rectifier. If the operating parameters of the converter are changed and the battery acts as a source supplying power to the AC system, the converter is then classified as an inverter.

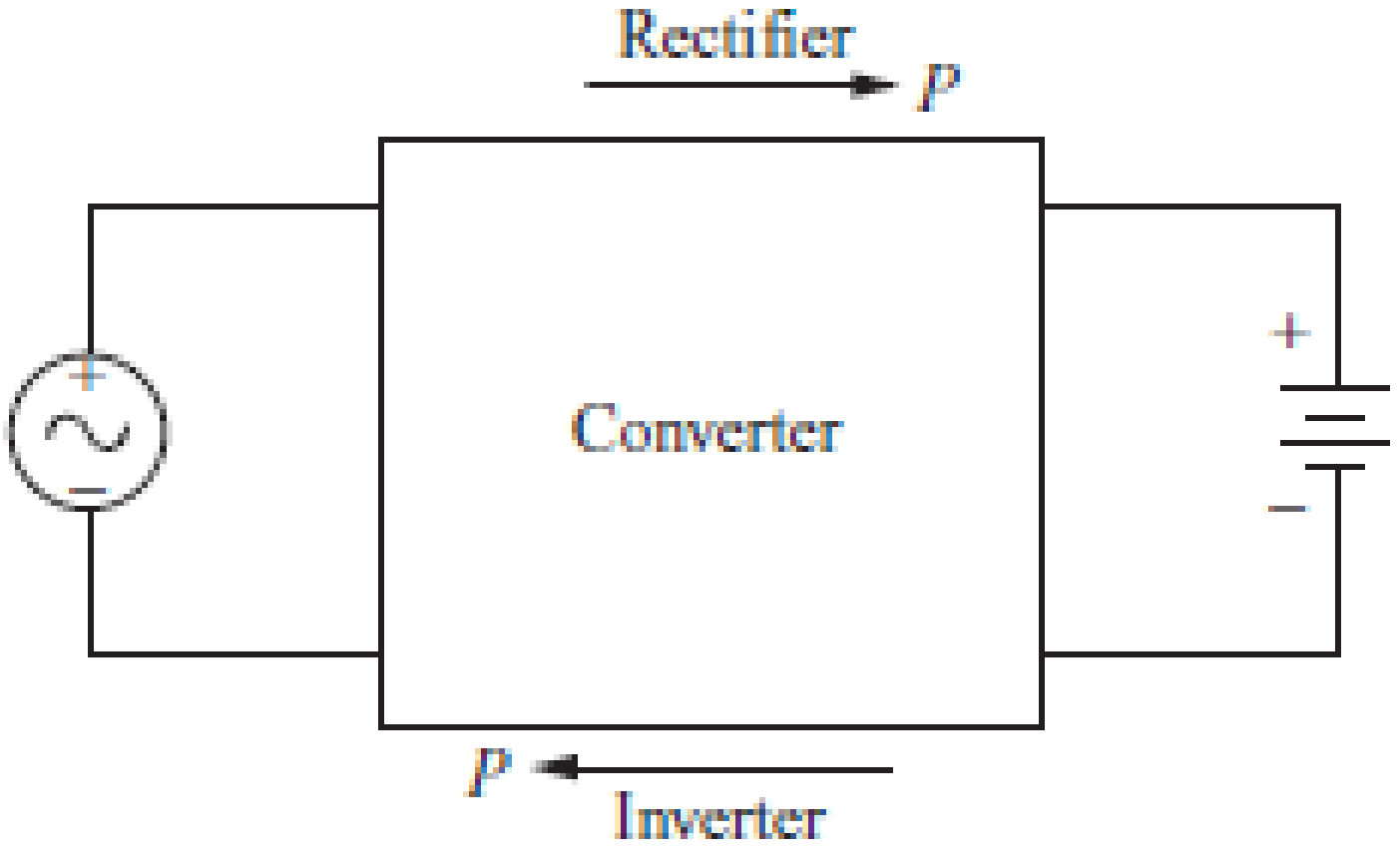


Figure 1-2 A converter can operate as a rectifier or an inverter, depending on the direction of average power *P*.

Power conversion can be a multistep process involving more than one type of converter.
For example, an AC-DC-AC conversion can be used to modify an AC source by first converting it to direct current and then converting the DC signal to an AC signal that has an amplitude and frequency different from those of the original AC source, as illustrated in Fig. 1-3.



Figure 1-3 two converters are used in a multistep process.

Power Electronics Concepts

- To illustrate some concepts in power electronics, consider the design problem of creating a 3-V DC voltage level from a 9-V battery.
- The purpose is to supply 3 V to a load resistance.
- One simple solution is to use a voltage divider, as shown in Fig. 1-4. For a load resistor R_L, inserting a series resistance of $2\,R_L$ results in $3\,V$ across R_L .
- A problem with this solution is that the power absorbed by the $2\,R_L$ resistor is twice as much as delivered to the load and is lost as heat, making the circuit only 33.3 percent efficient.

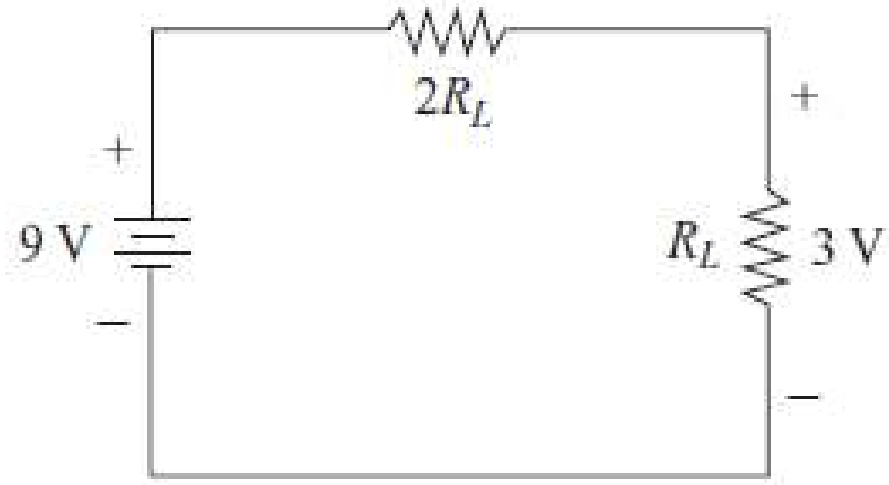


Figure 1-4 A simple voltage divider for creating 3 V from a 9-V source.

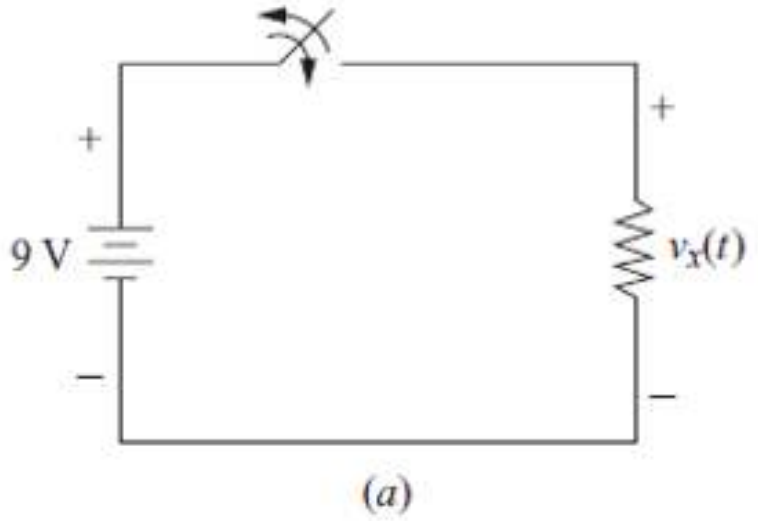
Another problem is that if the value of the load resistance changes, the output voltage will change unless the $2\,R_L$ resistance changes proportionally.

A solution to that problem could be to use a transistor in place of the 2 R_L resistance.

The transistor would be controlled such that the voltage across it is maintained at 6 V, thus regulating the output at 3 V.

However, the same low-efficiency problem is encountered with this solution.

To arrive at a more desirable design solution, consider the circuit in Fig. 1-5 *a*.



(*a*)

In this circuit, a switch is opened and closed periodically. The switch is a short circuit when it is closed and an open circuit when it is open, making the voltage across R_L equal to 9 V when the switch is closed and 0 V when the switch is open.

The resulting voltage across R_L will be like that of Fig. 1-5 b. This voltage is obviously not a constant DC voltage, but if the switch is closed for one-third of the period, the average value of v_{ave} (denoted as V_x) is one-third of the source voltage.

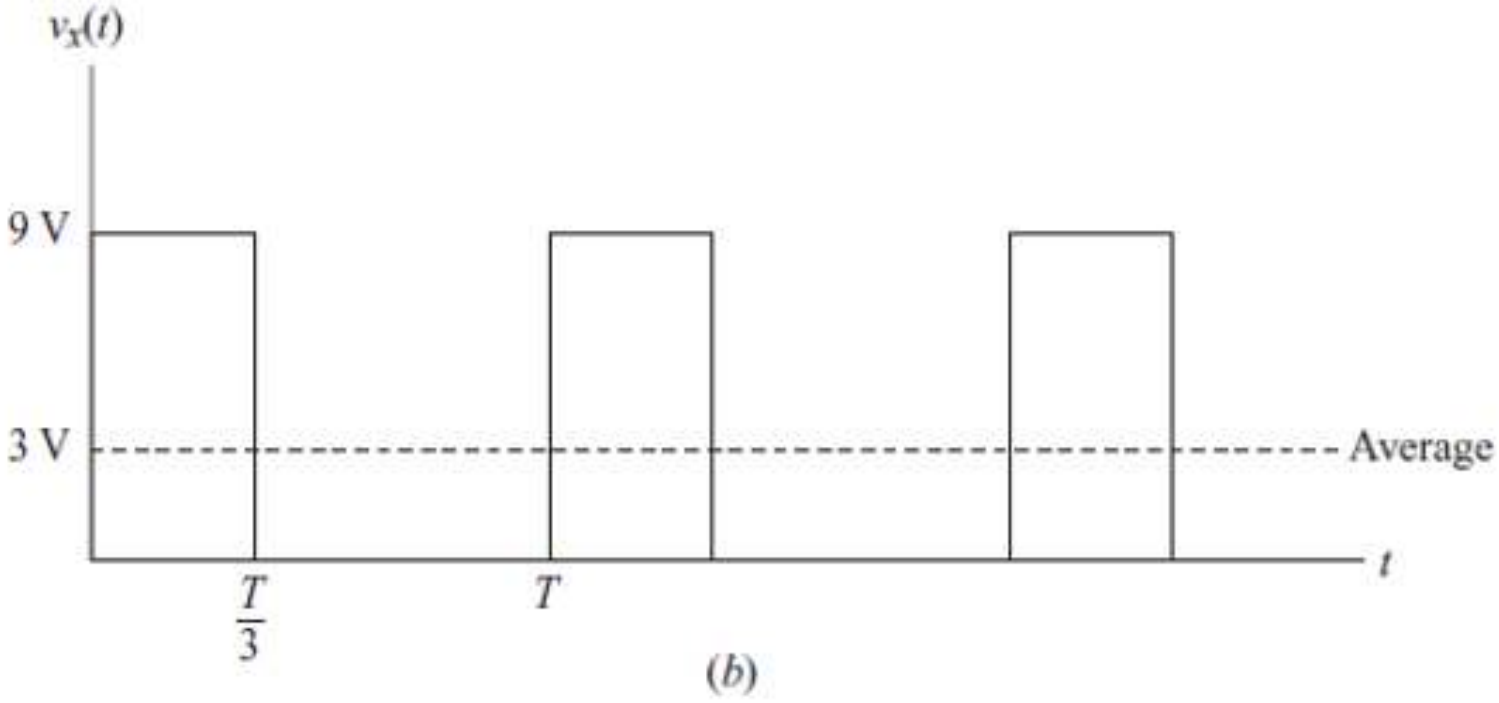


Figure 1-5 (*a*) A switched circuit; (*b*) a pulsed voltage waveform.

The average value is computed from the equation:

$$\text{avg}(v_x) = V_x = \frac{1}{T}\int_0^T v_x(t)\,dt = \frac{1}{T}\int_0^{T/3} 9\,dt + \frac{1}{T}\int_{T/3}^{T} 0\,dt = 3\text{ V} \qquad (1\text{-}1)$$

Considering the efficiency of the circuit, the instantaneous power absorbed by the switch is the product of voltage and current. When the switch is open, the power absorbed by it is zero because the current in it is zero. When the switch is closed, the power absorbed by it is zero because the voltage across it is zero. Since power absorbed by the switch is zero for both open and closed conditions, all power supplied by the 9-V source is delivered to R_L, making the circuit 100 percent efficient.

The circuit so far does not accomplish the design object of creating a DC voltage of 3 V. However, the voltage waveform v_x can be expressed as a Fourier series containing a DC term (the average value) plus sinusoidal terms at frequencies that are multiples of the pulse frequency.

To create a 3-V DC voltage, v_x is applied to a low-pass filter. An ideal low-pass filter allows the DC component of voltage to pass through to the output while removing the AC terms, thus creating the desired DC output. If the filter is zero losses, the converter will be 100 percent efficient.

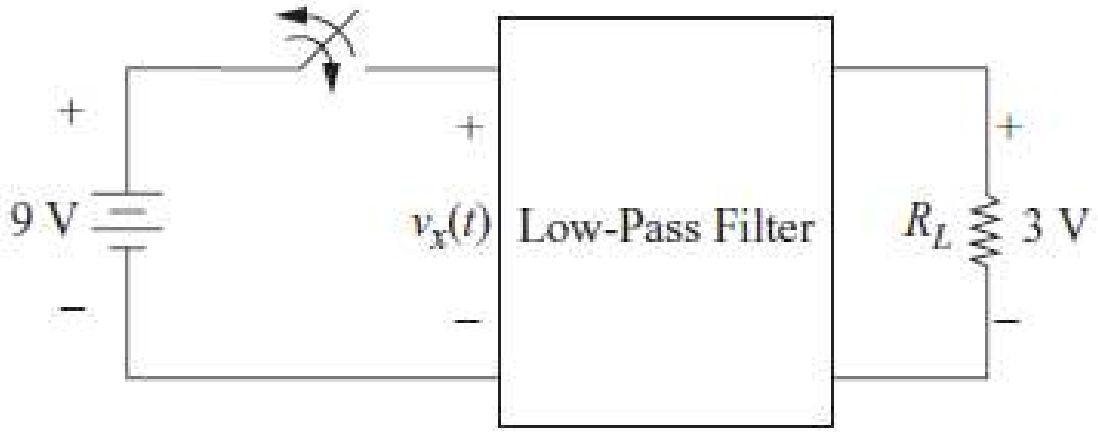


Figure 1-6 A low-pass filter allows just the average value of v_x to pass through to the load.

In practice, the filter will have some losses and will absorb some power. Additionally, the electronic device used for the switch will not be perfect and will have losses. However, the efficiency of the converter can still be quite high (more than 90 percent). The required values of the filter components can be made smaller with higher switching frequencies, making large switching frequencies desirable.

The “switch” in this example will be some electronic device, such as a metal-oxide field-effect transistor (MOSFET), or it may be comprised of more than one electronic device.

The power conversion process usually involves system control. Converter output quantities such as voltage and current are measured, and operating parameters are adjusted to maintain the desired output. For example, if the 9-V battery in the example in Fig. 1-6 decreased to 6 V, the switch would have to be closed 50 percent of the time to maintain an average value of 3 V for v_x.

A feedback control system would detect if the output voltage were not 3 V and adjust the closing and opening of the switch accordingly, as illustrated in Fig. 1-7.

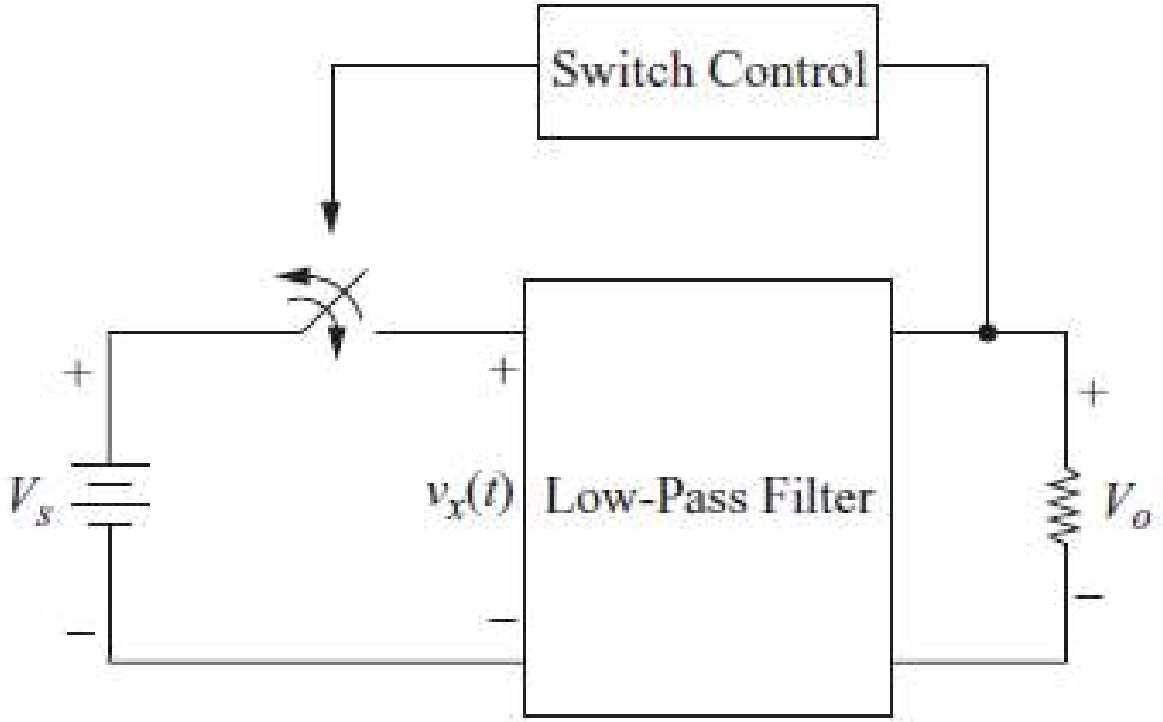


Figure 1-7 Feedback is used to control the switch and maintain the desired output voltage.

Electronic Switches

- An electronic switch is characterized by having two states: *on* and *off,* ideally being either a short circuit or an open circuit.
- Applications using switching devices are desirable because of the relatively small power loss in the device.
- If the switch is ideal, either the switch voltage or the switch current is zero, making the power absorbed by it zero.
- Real devices absorb some power when in the on state and when making transitions between the on and off states, but circuit efficiencies can still be quite high.
- Some electronic devices, such as transistors, can also operate in the active range where both voltage and current are not zero, but it is desirable to use these devices as switches when processing power.

The emphasis of this nots is on basic circuit operation rather than on device performance.

The particular switching device used in a power electronics circuit depends on the existing state of device technology.

The behaviors of power electronics circuits are often not affected significantly by the actual device used for switching, particularly if voltage drops across a conducting switch are small compared to other circuit voltages. Therefore, semiconductor devices are usually modeled as ideal switches so that circuit behavior can be emphasized.

Switches are modeled as short circuits when on and open circuits when off.

Transitions between states are usually assumed to be instantaneous, but the effects of non-ideal switching are discussed where appropriate.

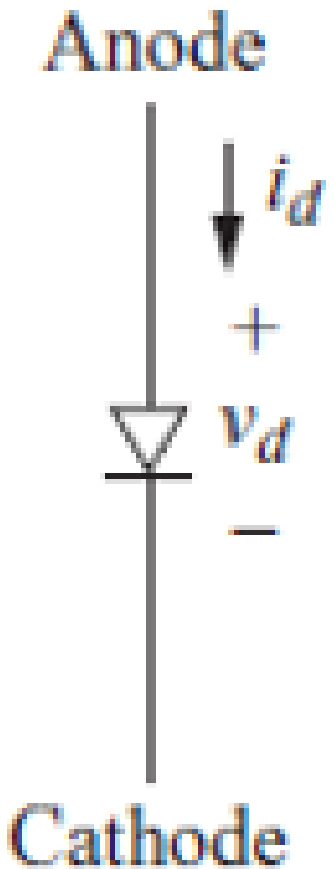
The Diode

A diode is the simplest electronic switch.

It is uncontrollable in that the on and off conditions are determined by voltages and currents in the circuit.

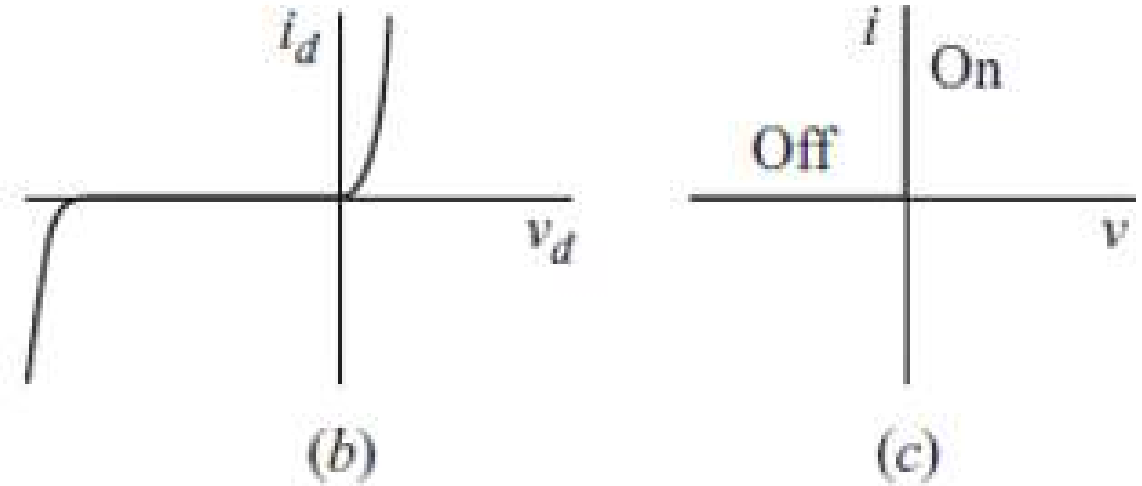
The diode is forward-biased (on) when the current i_d (Fig. 1-8 *a*) is positive and reverse-biased (off) when v_d is negative.

In the ideal case, the diode is a short circuit when it is forward-biased and is an open circuit when reverse-biased.



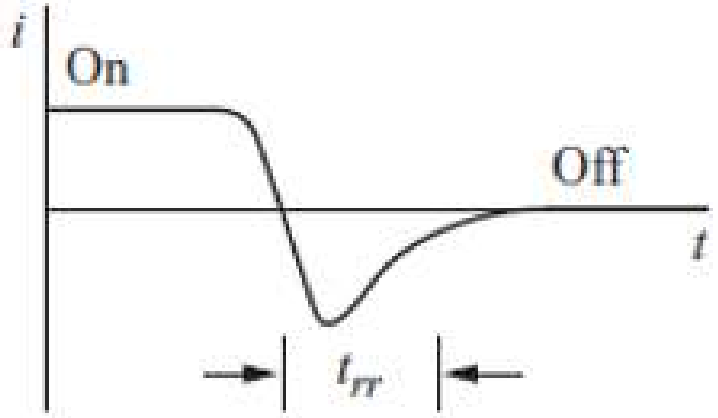
(*a*)

The actual and idealized current-voltage characteristics are shown in Fig. 1-8*b* and *c*. The idealized characteristic is used in most analyses in this text.



An important dynamic characteristic of a non-ideal diode is the reverse recovery current. When a diode turns off, the current in it decreases and momentarily becomes negative before becoming zero, as shown in Fig. 1-8*d*.

The time t_{rr} is the reverse recovery time, which is usually less than 1 μs. This phenomenon may become important in high-frequency applications. Fast-recovery diodes are designed to have a smaller t_{rr} than diodes designed for line-frequency applications



(*d*)

Silicon carbide (SiC) diodes have very little reverse recovery, resulting in more efficient circuits, especially in high-power applications.

Schottky diodes (Fig. 1-8*e*) have a metal-to-silicon barrier rather than a P-N junction.
Schottky diodes have a forward voltage drop of typically 0.3 V. These are often used in low-voltage applications where diode drops are significant relative to other circuit voltages. The reverse voltage for a Schottky diode is limited to about 100 V. The metal-silicon barrier in a Schottky diode is not subject to recovery transients and turn-on and off faster than P-N junction diodes.



(*e*)

Figure 1-8 (*a*) Rectifier diode; (*b*) *i-v* characteristic; (c) idealized *i-v* characteristic; (*d*) reverse recovery time t_{rr} ; (*e*) Schottky diode.

Thyristors

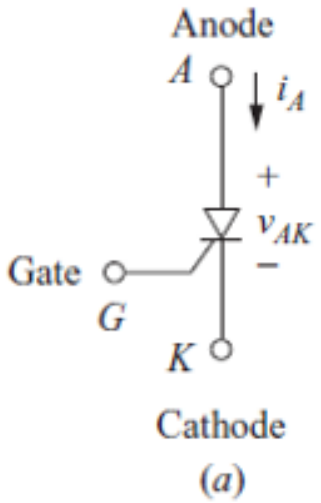
Thyristors are electronic switches used in some power electronic circuits where control of switch turn-on is required. The term *thyristor* often refers to a family of three-terminal devices that includes the silicon-controlled rectifier (SCR), the triac, the gate turnoff thyristor (GTO), the MOS-controlled thyristor (MCT), and others.

Thyristor and *SCR* are terms that are sometimes used synonymously.

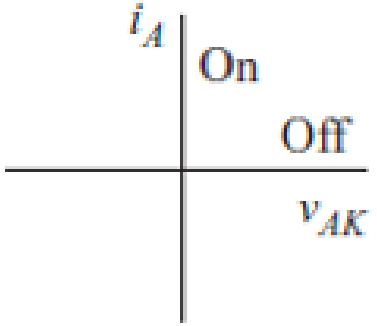
The SCR is the device used in this notes to illustrate controlled turn-on devices in the thyristor family.

Thyristors are capable of large currents and large blocking voltages for use in high-power applications, but switching frequencies cannot be as high as when using other devices such as MOSFETs.

The three terminals of the SCR are the anode, cathode, and gate (Fig.1-9*a*).



(*a*)



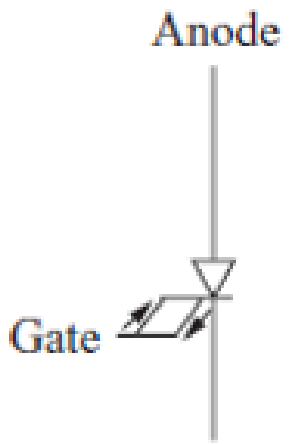
(*b*)

For the SCR to begin to conduct, it must have a gate current applied while it has a positive anode-to-cathode voltage. After conduction is established, the gate signal is no longer required to maintain anode current. The SCR will continue to conduct as long as the anode current remains positive and above a minimum value called the holding level.

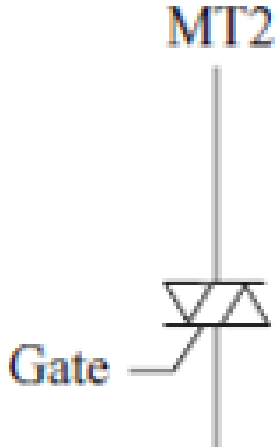
Figs. 1-9*a* and *b* show the SCR circuit symbol and the idealized current-voltage characteristic.

The gate turnoff thyristor (GTO) of Fig. 1-9*c*, like the SCR, is turned on by a short-duration gate current if the anode-to-cathode voltage is positive. However, unlike the SCR, the GTO can be turned off with a negative gate current.

The GTO is therefore suitable for some applications where control of both turn-on and turn-off of a switch is required. The negative gate turn-off current can be of brief duration (a few microseconds), but its magnitude must be very large compared to the turn-on current. Typically, gate turnoff current is one-third the on-state anode current. The idealized *i*-*v* characteristic is like that of Fig. 1-9*b* for the SCR.



(*c*)



(*d*)

The triac (Fig. 1-9 *d*) is a thyristor that can conduct current in either direction. The triac is functionally equivalent to two antiparallel SCRs (in parallel but in opposite directions). Common incandescent light-dimmer circuits use a triac to modify both the positive and negative half cycles of the input sine wave.

The MOS-controlled thyristor (MCT) in Fig. 1-9*e* is functionally equivalent to a GTO but without the high turn-off gate current requirement. The MCT has an SCR and two MOSFETs integrated into one device. One MOSFET turns the SCR on, and one MOSFET turns the SCR off. The MCT is turned on and off by establishing the proper voltage from gate to cathode, as opposed to establishing a gate current in the GTO.

Thyristors were historically the power electronics switch of choice because of high voltage and current ratings available. Thyristors are still used, especially in high-power applications, but ratings of power transistors have increased greatly, making the transistor more desirable in many applications.

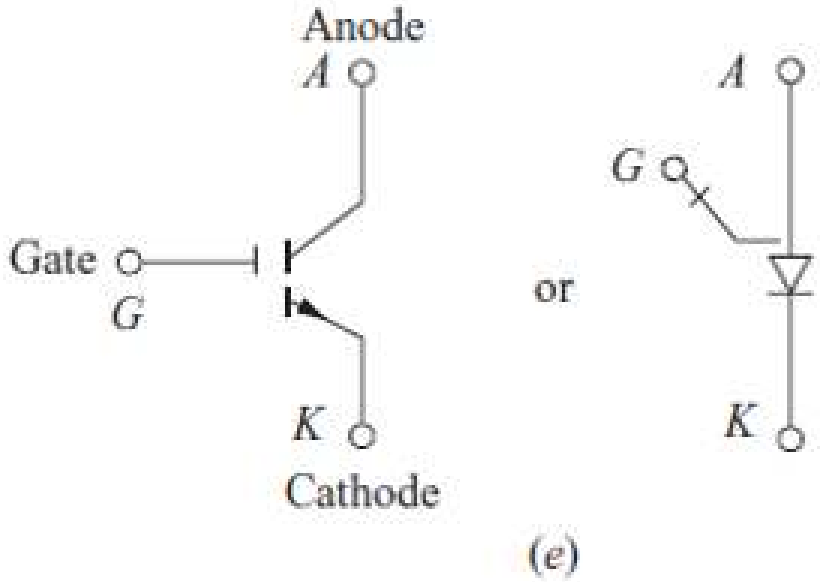
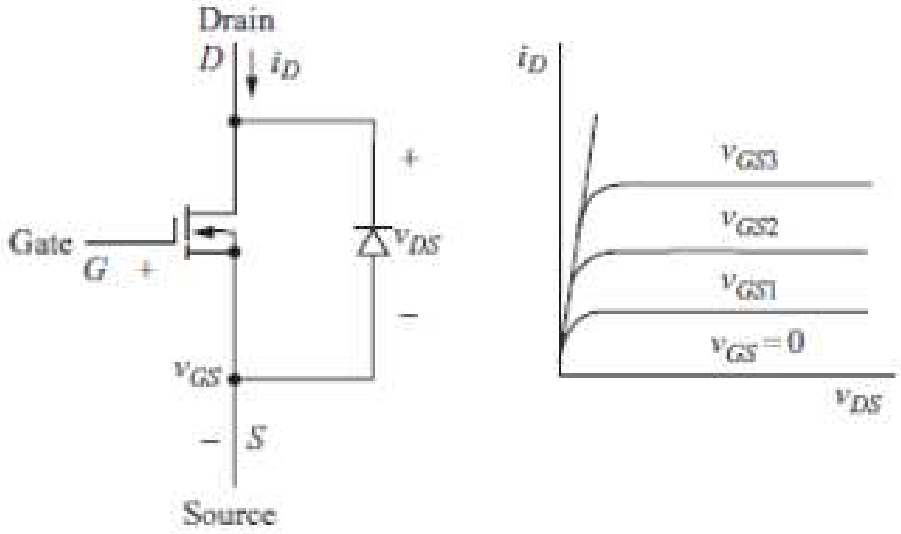


Figure 1-9 Thyristor devices: (*a*) silicon-controlled rectifier (SCR); (*b*) SCR idealized *i*-*v* Characteristic; (*c*) gate turnoff (GTO) thyristor; (*d*) triac; (*e*) MOS-controlled thyristor (MCT).

Transistors

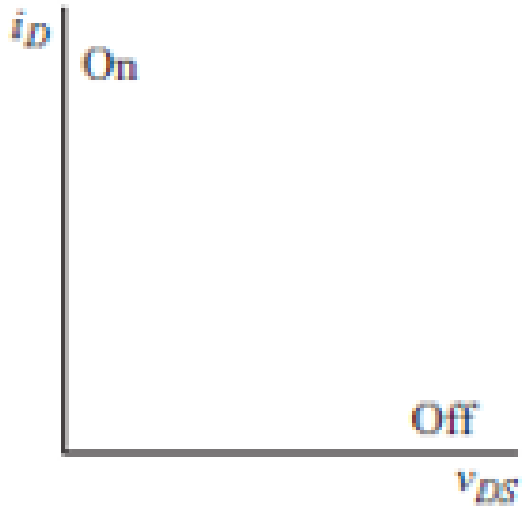
- Transistors are operated as switches in power electronics circuits. Transistor drive circuits are designed to have the transistor either in the fully on or fully off state.
- This differs from other transistor applications, such as in a linear amplifier circuit, where the transistor operates in a region having simultaneously high voltage and current.
- Unlike the diode, turn-on and turnoff of a transistor are controllable. Types of transistors used in power electronics circuits include MOSFETs, bipolar junction transistors (BJTs), and hybrid devices such as insulated-gate bipolar junction transistors (IGBTs). Figs. 1-10 to 1-12 show the circuit symbols and the current-voltage characteristics.

The MOSFET (Fig. 1-10*a*) is a voltage-controlled device with characteristics as shown in Fig. 1-10*b*. MOSFET construction produces a parasitic (body) diode, as shown, which can sometimes be used to an advantage in power electronics circuits. Power MOSFETs are of the enhancement type rather than the depletion type.



A sufficiently large gate-to-source voltage will turn the device on, resulting in a small drain-to-source voltage. In the on state, the change in v_{DS} is linearly proportional to the change in i_D. Therefore, the on MOSFET can be modeled as an on-state resistance called $R_{DS(on)}$.

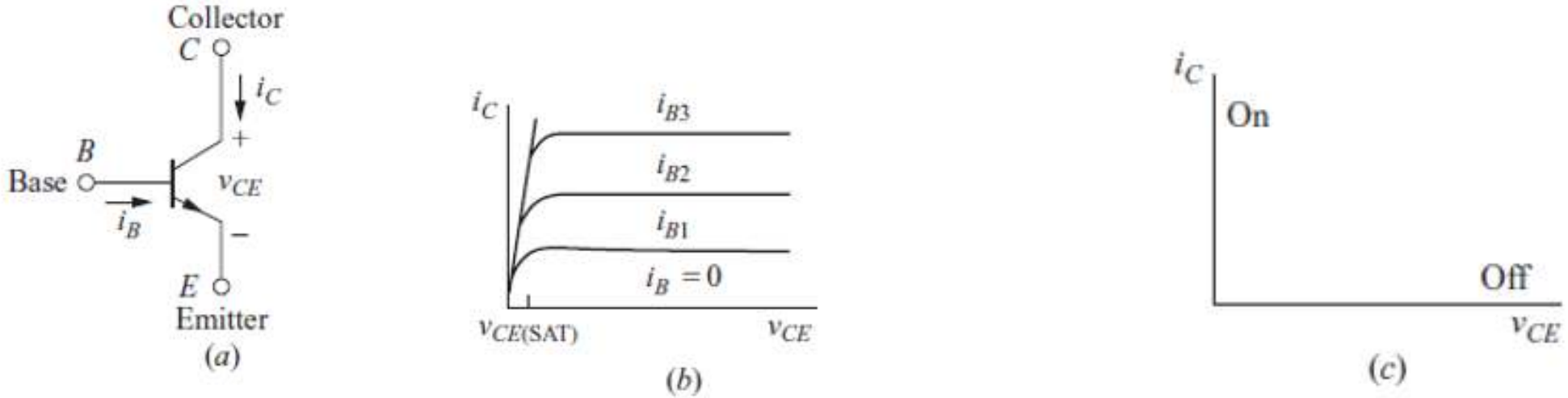
MOSFETs have on-state resistances as low as a few milliohms. For a first approximation, the MOSFET can be modeled as an ideal switch with a characteristic shown in Fig. 1-10c. Ratings are to 1500 V and more than 600 A (although not simultaneously). MOSFET switching speeds are greater than those of BJTs and are used in converters operating into the megahertz range.



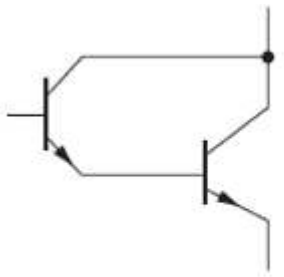
(c)

Figure 1-10 (*a*) MOSFET (N-channel) with body diode; (*b*) MOSFET characteristics; (*c*) idealized MOSFET characteristics.

Typical BJT characteristics are shown in Fig. 1-11*b*. The on state for the transistor is achieved by providing sufficient base current to drive the BJT into saturation. The collector-emitter saturation voltage is typically 1 to 2 V for a power BJT. Zero base current results in an off transistor. The idealized *i*-*v* characteristic for the BJT is shown in Fig. 1-11*c*.



The BJT is a current-controlled device, and power BJTs typically have low h_{FE} values, sometimes lower than 20. If a power BJT with $h_{FE} = 20$ is to carry a collector current of 60 A, for example, the base current would need to be more than 3 A to put the transistor into saturation. The drive circuit to provide a high base current is a significant power circuit in itself. Darlington configurations have two BJTs connected as shown in Fig. 1-11*d*.



(*d*)

Figure 1-11 (*a*) BJT (NPN); (*b*) BJT characteristics; (*c*) idealized BJT characteristics; (*d*) Darlington configuration.

The effective current gain of the combination is approximately the product of individual gains and can thus reduce the current required from the drive circuit. The Darlington configuration can be constructed from two discrete transistors or can be obtained as a single integrated device. Power BJTs are rarely used in new applications, being surpassed by MOSFETs and IGBTs.

The IGBT of Fig. 1-12 is an integrated connection of a MOSFET and a BJT. The drive circuit for the IGBT is like that of the MOSFET, while the on-state characteristics are like those of the BJT. IGBTs have replaced BJTs in many applications.



Figure 1-12 IGBT: (*a*) Equivalent circuit; (*b*) circuit symbols.

Switch Selection

- The selection of a power device for a particular application depends not only on the required voltage and current levels but also on its switching characteristics.
- Transistors and GTOs provide control of both turn-on and turn-off, SCRs of turn-on but not turn-off, and diodes of neither.
- Switching speeds and the associated power losses are very important in power electronics circuits. The BJT is a minority carrier device, whereas the
- MOSFET is a majority carrier device that does not have minority carrier storage delays, giving the MOSFET an advantage in switching speeds. BJT switching times may be a magnitude larger than those for the MOSFET. Therefore, the MOSFET generally has lower switching losses and is preferred over the BJT.
- When selecting a suitable switching device, the first consideration is the required operating point and turn-on and turn-off characteristics. Example 1-1 outlines the selection procedure.

Example 1-1 (Switch Selection)

- The circuit of Fig. 1-13*a* has two switches. Switch S_1 is on and connects the voltage source ($V_S = 24\ V$) to the current source ($I_O = 2\ A$). It is desired to open switch S_1 to disconnect V_S from the current source. This requires that a second switch S_2 close to provide a path for current I_O, as in Fig. 1-13*b*. At a later time, S_1 must reclose and S_2 must open to restore the circuit to its original condition. The cycle is to repeat at a frequency of 200 kHz.
- Determine the type of device required for each switch and the maximum voltage and current requirements of each.

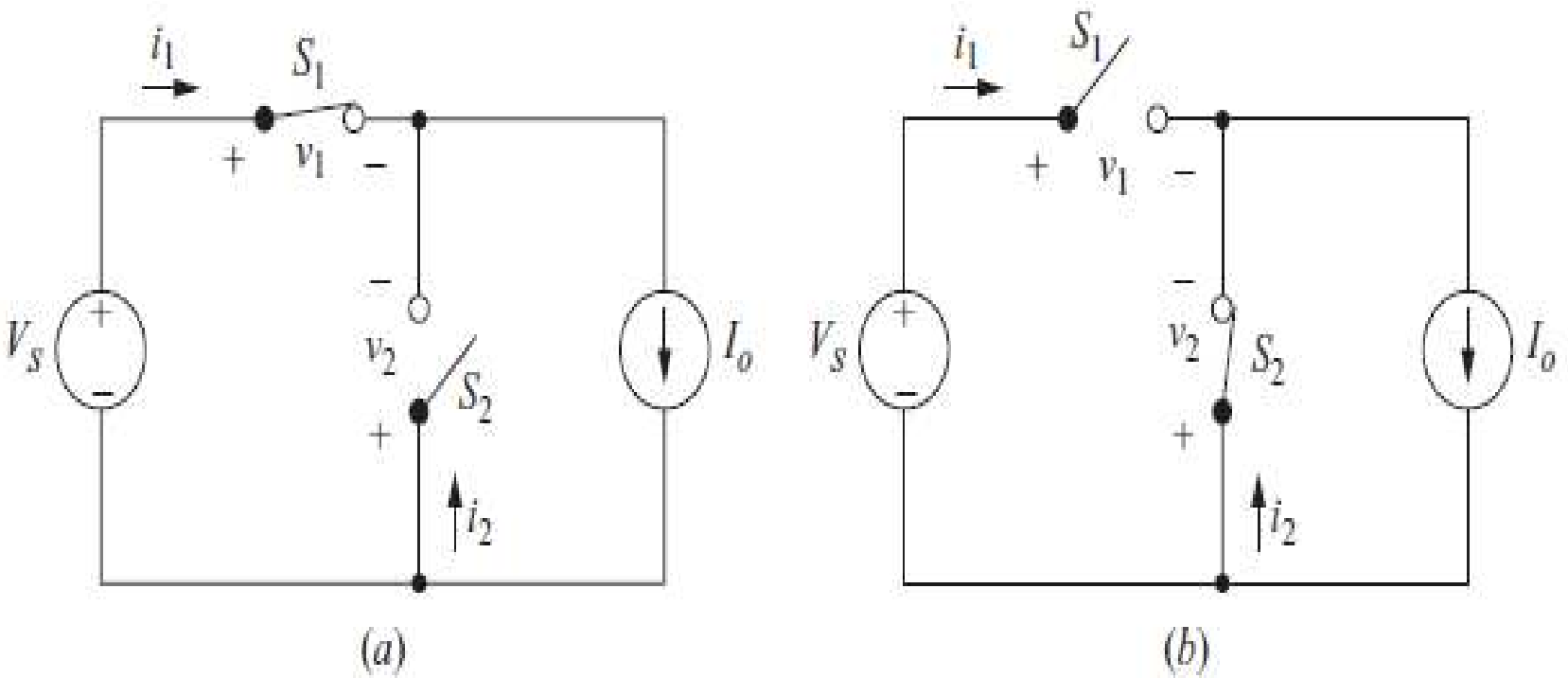


Figure 1-13 Circuit for Example 1-1. (*a*) S_1 closed, S_2 open; (*b*) S_1 open, S_2 closed.

■ Solution

- The type of device is chosen from the turn-on and turn-off requirements, the voltage and current requirements of the switch for the on and off states, and the required switching speed.
- The steady-state operating points for S_1 are at $(v_1, i_1) = (0, I_O)$ for S_1 closed and $(V_S, 0)$ for the switch open (Fig. 1-13c).

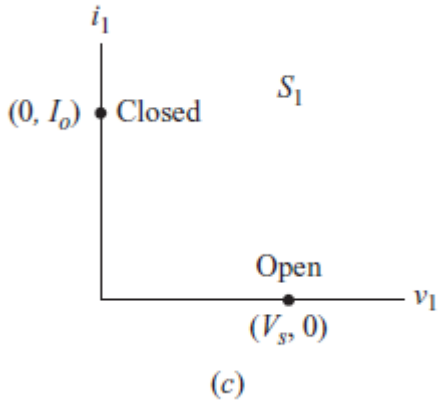


Figure 1-13 Circuit for Example 1: (c) operating points for S_1

The operating points are on the positive i and v axes, and S_1 must turn off when $i_1 = I_O > 0$ and must turn on when $v_1 = V_S > 0$. The device used for S_1 must therefore provide control of both turn-on and turnoff.

The MOSFET characteristic of Fig. 1-10d or the BJT characteristic of Fig. 1-11c matches the requirement. A MOSFET would be a good choice because of the required switching frequency, simple gate-drive requirements, and relatively low voltage and current requirement (24 V and 2 A).

The steady-state operating points for S_2 are at $(v_2, i_2) = (-V_S, 0)$ in Fig. 1-13a and $(0, I_O)$ in Fig. 1-13b, as shown in Fig. 1-13d.

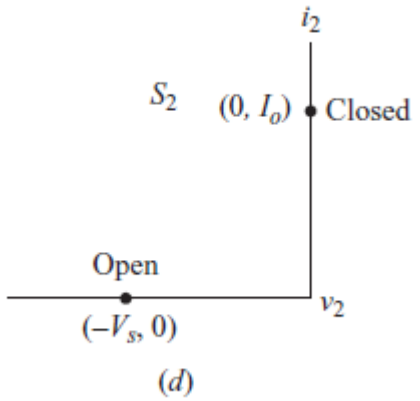


Figure 1-13 Circuit for Example 1: (*d*) operating points for S_2.

The operating points are on the positive current axis and negative voltage axis. Therefore, a positive current in S_2 is the requirement to turn S_2 on, and a negative voltage exists when S_2 must turn off. Since the operating points match the diode (Fig. 1-8*c*) and no other control is needed for the device, a diode is an appropriate choice for S_2.

Figure 1-13*e* shows the implementation of the switching circuit. Maximum current is 2 A, and maximum voltage in the blocking state is 24 V.

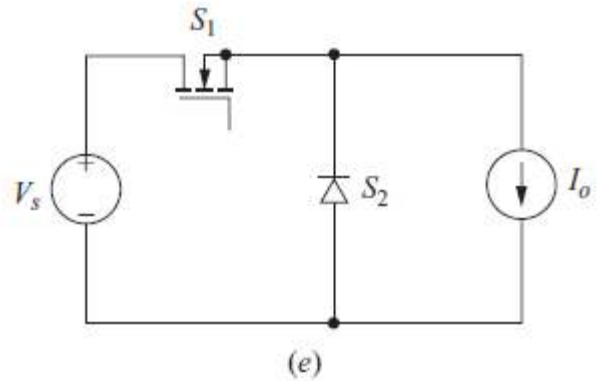


Figure 1-13 Circuit for Example 1: (*e*) switch implementation using a MOSFET and diode.

Although a diode is a sufficient and appropriate device for the switch S_2, a MOSFET would also work in this position, as shown in Fig. 1-13*f*. When S_2 is on and S_1 is off, current flows upward out of the drain of S_2.

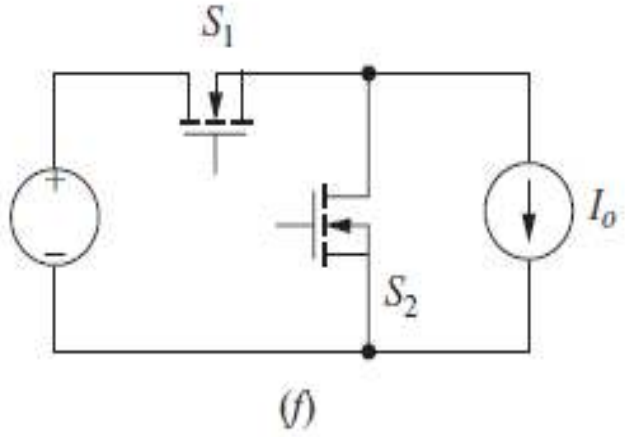


Figure 1-13 Circuit for Example 1: (*f*) switch implementation using two MOSFETs (synchronous rectification).

The advantage of using a MOSFET is that it has a much lower voltage drop across it when conducting compared to a diode, resulting in lower power loss and a higher circuit efficiency. The disadvantage is that a more complex control circuit is required to turn on S_2 when S_1 is turned off. However, several control circuits are available to do this. This control scheme is known as synchronous rectification or synchronous switching.

In a power electronics application, the current source in this circuit could represent an inductor that has a nearly constant current in it.